TAEYOUNG YUN

Ph.D student @ KAIST

99yty@kaist.ac.kr | Taeyoung Yun

dbsxodud-11 | dbsxodud-11.github.io

RESEARCH INTEREST

My research interest lies in building generative agents that make reliable decisions in the real world. Simulators of real-world systems are becoming easier to build with the help of LLMs, but they never perfectly reflect reality or user intent. I am interested in how to align simulators with user intent and how to train generative policies that stay robust under such uncertainty. To this end, my research focuses on user-simulator alignment, reinforcement learning under uncertainty, and black-box optimization.

During my earlier PhD years, I studied amortized samplers, such as GFlowNets, that learn diverse generative policies for text, image, and biological sequence generation. I also worked on transportation-related projects, such as traffic light management and network layout design, where I formulated each task as a black-box optimization problem over an existing simulator. My current research topics are:

- **RL for Scientific Discovery:** Developing an off-policy RL algorithm to search for high-performing and diverse candidates for scientific discovery **([C1], [C2], [C6], [C8], [C16], [P1], [P3], [P4])**.
- **Generative models for Optimization**: Developing a generative model-based approach (Diffusion / Flow-based models) for searching promising candidates in high-dimensional space **([C9], [C10], [C11], [C17])**.

EDUCATION

03/2024 - Current **Ph.D Student in Industrial and Systems Engineering** ***KAIST***

Supervised by Jinkyoo Park

08/2022 - 02/2024 **M.S in Graduate School of AI** ***KAIST***

Supervised by Jinkyoo Park

M.S Thesis: Offline Meta Black-box Optimization Framework for Intelligent Traffic Light Management System

03/2018 - 08/2022 **B.S in Industrial and Systems Engineering & Computer Science** ***KAIST***

INTERNSHIPS

08/2026 - 10/2026 **Research Intern at NaverLabs Europe** ***Grenoble, France***

Hosted by Darko Drakulic and Sofia Michel

Develop a scalable RL algorithm for online multi-agent coordination under uncertainty

06/2025 - 11/2025 **Research Intern at Mila** ***Montreal, Canada***

Hosted by Yoshua Bengio

RL-based red-teaming with evolving environments by safety fine-tuning the victim LLM to promote easy-to-hard exploration.

09/2024 - 03/2025 **Visiting Intern at HKUST** ***Hong Kong (Remote)***

Hosted by Ling Pan

Fine-tuning LLM with GFlowNets to generate diverse and effective prompts for text-to-image diffusion models.

03/2021 - 08/2021 **Research Intern at Kakao Recommendation Team** ***Seoul, Korea***

Develop contextual bandit algorithms for a personal recommendation.

Analyze the gap between simulation and real-world deployment.

INDUSTRIAL PROJECTS

03/2026 - Current **Agentic AI for Traffic-related Question Answering** ***Seoul, Korea***

Collaborate with GS

Develop an Agentic AI system for developing LLM-based for better traffic-related question answering framework.

09/2024 - 09/2025 **Traffic Network Layout Optimization** ***Seoul, Korea***

Collaborate with GS

Develop a Generative model-based design algorithm for optimizing traffic network layout on a given traffic pattern.

03/2023 - 03/2024 **Incentive Design for Managing Taxi Fleet** ***Daejeon, Korea***

Collaborate with ETRI

Develop an RL-based incentive design algorithm for rebalancing taxi fleets to resolve the taxi imbalance problem.

03/2022 - 03/2023 **Traffic Light Optimization** ***Seoul, Korea***

Collaborate with KT

Develop a Bayesian optimization algorithm for managing multiple traffic lights in the real world to reduce congestion.

PUBLICATIONS

*: Equal Contribution

Preprints

Arxiv, 2026 **[P4] Noise-Space Reward Alignment for Few-Step Generative Models**
Woocheol Shin, **Taeyoung Yun**, Jiwon Sung, Hyeongyu Kang, Jeongmin Park, Jaewoo Lee, Kyuil Sim, Dohyun Kim, Yutong He, and Jinkyoo Park
Paper / Code

Arxiv, 2026 **[P3] Decoupling Exploration and Exploitation in GFlowNets via Policy Evaluation**
Taeyoung Yun*, Inhyuck Song*, Jinkyoo Park, and Ling Pan
Paper / Code

Arxiv, 2026 (based on ICMLW) **[P2] Robust Exploration through Generative Replay**
Inhyuck Song, **Taeyoung Yun**, Jaewoo Lee, and Jinkyoo Park
Paper / Code

Arxiv, 2025 **[P1] Improving Sampling Distribution of Off-policy Training in Generative Flow Networks**
Taeyoung Yun, Sujin Yun, Jinkyoo Park, and Ling Pan
Paper / Code

Conference Publications

NeurIPS, 2026 (based on NIPSW) **[C17] Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization**
Kiyoung Om*, Kyuil Sim*, **Taeyoung Yun***, Hyeongyu Kang, and Jinkyoo Park
Paper / Code

NeurIPS, 2026 (based on ICMLW) **[C16] Aligning Few-Step Generative Models by Amortizing Sample-based Variational Inference**
Jaewoo Lee*, Hyeongyu Kang*, Dohyun Kim, Kyuil Sim, Woocheol Shin, Minsu Kim, **Taeyoung Yun**, Jeongjae Lee, Sanghyeok Choi, Tabitha Edith Lee, Jongchul Ye, Jinkyoo Park
Paper / Code

NeurIPS, 2026 (based on ICMLW)
[C15] Automated Kernel Discovery for Understanding High-dimensional Bayesian Optimization
Taeyoung Yun*, Woocheol Shin*, Inhyuck Song, Jaewoo Lee, and Jinkyoo Park
Paper / Code

CIKM, 2026 (Short Paper)
[C14] SUMO-SimQA: A Benchmark for LLM Cross-File Reasoning over Traffic Simulation Files
Jeeyun Chang*, **Taeyoung Yun***, Minwoo Jeong, Jinkyoo Park, and Yoonjin Yoon
Paper / Code

ICML, 2026
[C13] Active Attacks: Red-teaming LLMs via Adaptive Environments
Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim
Paper / Code

ICLR, 2026
[C12] Diffusion Alignment as Variational Expectation-Maximization
Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, **Taeyoung Yun**, Kiyoung Om, and Jinkyoo Park
Paper / Code

WSDM, 2026
[C11] Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models
Taeyoung Yun, Inhyuck Song, Woocheol Shin, Yujin Shin, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
Paper / Code

KDD, 2025
[C10] Wind Farm Layout Optimization with Diffusion Models
Yujin Shin*, **Taeyoung Yun***, Sujin Yun, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
Paper / Code

ICML, 2025 (based on ICLRW)
[C9] Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization
Taeyoung Yun*, Kiyoung Om*, Jaewoo Lee, Sujin Yun, and Jinkyoo Park
Paper / Code

ICML, 2025 (based on NIPSW)
[C8] Improved Off-Policy Reinforcement Learning in Biological Sequence Design
Hyeonah Kim, Minsu Kim, **Taeyoung Yun**, Sanghyeok Choi, Emmanuel Bengio, Alex Hernández Garcia, and Jinkyoo Park
Paper / Code

CVPR, 2025
[C7] Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation
Taeyoung Yun, Dinghuai Zhang, Jinkyoo Park, and Ling Pan
Paper / Code

ICLR, 2025
[C6] Adaptive Teachers for Amortized Samplers
Minsu Kim*, Sanghyeok Choi*, **Taeyoung Yun**, Emmanuel Bengio, Leo Feng, Jarrid Rector-Brooks, Sungsoo Ahn, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio
Paper / Code

NeurIPS, 2024
[C5] Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization
Taeyoung Yun, Sujin Yun, Jaewoo Lee, and Jinkyoo Park
Paper / Code

NeurIPS, 2024 (based on ICLRW)
[C4] GTA: Generative Trajetory Augmentation with Guidance for Offline Reinforcement Learning
Jaewoo Lee*, Sujin Yun*, **Taeyoung Yun**, and Jinkyoo Park
Paper / Code

KDD, 2024
[C3] An Offline Meta Black-box Optimization Framework for Adaptive Design of Urban Traffic Light Management Systems
Taeyoung Yun*, Kanghoon Lee*, Sujin Yun, Ilmyung Kim, Won-Woo Jung, Min-Cheol Kwon, Kyujin Choi, Yoohyeon Lee, and Jinkyoo Park
Paper / Code

ICML, 2024 (based on NIPSW)
[C2] Learning to Scale Logits for Temperature-conditional GFlowNets
Minsu Kim*, Juhwan Ko*, **Taeyoung Yun***, Dinghuai Zhang, Ling Pan, Woochang Kim, Jinkyoo Park, and Yoshua Bengio
Paper / Code

ICLR, 2024 (Spotlight)
[C1] Local Search GFlowNets
Minsu Kim, **Taeyoung Yun**, Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, Sungsoo Ahn, and Jinkyoo Park
Paper / Code

Journal Publications

TMLR, 2026 (Survey Certification)
[J1] Offline Model-Based Optimization: Comprehensive Review
Minsu Kim, Jiayao Gu, Ye Yuan, **Taeyoung Yun**, Zixuan Liu, Yoshua Bengio, Can Chen
Paper / Code

TEACHING EXPERIENCES

2026 **Teaching Assistant** ***KAIST***
IE579: Game Theory and Multi-Agent Reinforcement Learning

2025 **Teaching Assistant** ***KAIST***
IE343: Statistical Machine Learning
IE437: Data-Driven Decision Making and Control

2024 **Teaching Assistant** ***KAIST***
IE437: Data-Driven Decision Making and Control
IE481: Manufacturing & Artificial Intelligence

2022 **Teaching Assistant** ***KAIST***
MAS480: Introduction to Scientific Machine Learning

ACADEMIC SERVICES

2026 **Reviewer**
AAAI, KDD, ICLR, AAMAS, AISTATS, TMLR, TPAMI, ICML, RLC, NeurIPS

2025 **Reviewer**
ICLR, AAMAS, AISTATS, ICML, KDD, TMLR, NeurIPS

INVITED TALKS

2026 **AutoML Seminar**
Posterior Inference in Generative Models for High-dimensional Black-box Optimization

HONORS & AWARDS

2025 **Qualcomm Innovation Fellowship Finalist** ***Seoul, Korea***
Paper: Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation

2025 **Top Reviewer** ***NeurIPS***
Honor for Top 8% Reviewers

2021 **Dean's List** ***KAIST***

Honor for Top 2% Students

2021 **Excellence Award (2nd Place)** ***Seoul, Korea***

Big Data Competition Hosted by NH